6. Differentiation

What students need to learn:

Differentiation of e^x, $\ln x$, $\sin x$, $\cos x$, $\tan x$ and their sums and differences.	
Differentiation using the product rule, the quotient rule and the chain rule.	Differentiation of $\operatorname{cosec} x$, $\cot x$ and $\sec x$ are required. Skill will be expected in the differentiation of functions generated from standard forms using products, quotients and composition, such as $2x^4 \sin x$, $\dfrac{e^{3x}}{x}$, $\cos x^2$ and $\tan^2 2x$.
The use of $\dfrac{dy}{dx} = \dfrac{1}{\left(\dfrac{dx}{dy}\right)}$.	E.g. finding $\dfrac{dy}{dx}$ for $x = \sin 3y$.
Differentiation of simple functions defined implicitly or parametrically.	The finding of equations of tangents and normals to curves given parametrically or implicitly is required.
Exponential growth and decay.	Knowledge and use of the result $\dfrac{d}{dx}(a^x) = a^x \ln a$ is expected.
Formation of simple differential equations.	Questions involving connected rates of change may be set.